**Indian Institute of Technology, Kanpur**

**ECO731: Public Economics and Public**

**Policy**

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**First Course Handout**

1. **Contact Information:** ndatta@iitk.ac.in.
2. **Office Hours:** To be decided.
3. **Lectures:** Mondays and Thursdays, 2 PM– 3:15 PM, Room L13.
4. **Pre-requisites:** None.

5. **Course Description:**

This course provides an advanced exploration of the theoretical foundations of public economics, with an emphasis on understanding the fundamental trade-offs and welfare implications of government interventions in market outcomes. Major topics covered include static and dynamic optimal taxation, public goods, externalities, fiscal federalism, public debt, and dynamic public policy. It also delves into the public finance system in India, with an emphasis on key policy measures such as the Fiscal Responsibility and Budget Management Act, and the Goods and Services Tax framework. While the primary focus will be on the rigorous mathematical examination of established theoretical frameworks in the literature, relevant empirical papers will also be covered to provide a comprehensive understanding of critical issues in public economics.

**a. Objectives:**

Upon completion of this course, students should be able to:

1. Demonstrate a deep understanding of the theoretical foundations of public economics and analyze the welfare implications of government interventions in market outcomes.
2. Apply theoretical models and empirical evidence to analyze key public policy measures.
3. Develop the ability to read, understand, and critique peer-reviewed research articles in the field, and apply this knowledge to formulate original research.

**b. Tentative Course Schedule:**

| S. No. | Broad Title | Topics | No. of Lectures |
|---|---|---|---|
| **1.** | Static Optimal Taxation (***SOT***) | **Lecture 1** - optimal commodity taxation; many-person Ramsey tax rule<br><br>**Lecture 2** - production efficiency theorem<br><br>**Lecture 3** - non-linear income taxation | 3 |
| **2.** | Public Goods (***PG***) | **Lecture 4** - neoclassical theory of public goods; Samuelson Condition; market failure; Lindahl equilibrium<br><br>**Lecture 5** - mechanism design and Vickery-Clarke-Groves mechanism, median voter theorem, public good provision with distortionary taxation; Laffer curve | 2 |
| **3.** | Externalities (***E***) | **Lecture 6** - externalities and market failure; Pigouvian taxation; Coase Theorem<br><br>**Lecture 7** - price versus quantity regulation; tradeable permits; optimal commodity taxation with externalities | 2 |

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| **4.** | Dynamic Optimal Taxation (***DOT***) | **Lecture 8 -** Ricardian Equivalence<br><br>**Lecture 9 -** the tax-smoothing problem<br><br>**Lecture 10 -** optimal capital taxation<br><br>**Lecture 11 -** time-inconsistencyin government policy; New Dynamic Public Finance | 4 |
| **5.** | Optimal Transfer Programs (***OTP***) | **Lecture 12 -** optimal design of cash and in-kind transfers<br><br>**Lecture 13 -** transfers as insurance policy; Samaritan's Dilemma | 2 |
| **6.** | Fiscal Federalism (***FF***) | **Lecture 14 -** task assignment and Oates' decentralization theorem; inter-jurisdictional tax competition<br><br>**Lecture 15 -** vertical fiscal imbalances; intergovernmental transfers in India | 2 |
| **7.** | Revenue and Expenditure in India (***REI***) | **Lecture 16 -** fiscal assignment; components of expenditure; tax devolution; role of the Finance Commission<br><br>**Lecture 17 -** direct and indirect taxes; the Goods and Services Tax framework | 2 |
| **8.** | Public Debt in India (***PDI***) | **Lecture 18 -** deficit financing; inflation and debt dynamics<br><br>**Lecture 19 -** fiscal rules; Fiscal Responsibility and Budget Management Act (2003) | 2 |
| **9.** | Economics, Politics, and Policy Choice (***E&P***) | **Lecture 20 -** Probabilistic Voting and income redistribution<br><br>**Lecture 21 -** legislative bargaining<br><br>**Lecture 22 -** electoral incentives and public good provision | 3 |
| **10.** | Dynamic Public Policy (***DPP***) | **Lecture 23 -** dynamic policyproblems<br><br>**Lectures 24&25 -** political economy of taxation, spending, and public debt | 3 |

## 6. Grading Policy:

The total grades from the course will be distributed among three components. The points will be allocated as follows:

a. **Referee Report (20%) –** A list of papers will be provided. Students will choose one paper (on a first come first served basis) and write a report on it. This will be due near the end of the midterm exams week (a specific date will be provided in class). The structure of the report should be the following-
   i. **Recommendation** (1-2 sentences) – Are you recommending a rejection, an acceptance, or a revise & resubmit?
   ii. **Critical overview** (approximately 500 words) – What is the paper's contribution to the literature, and how significant is it in your opinion? Note that this will be more than just a summary of the paper. If there are specific theoretical, empirical, or computational innovations, then mention those.
   iii. **Analysis** (approximately 1000 words) – What are the strengths and weaknesses of the paper? In the theory, what are the main assumptions driving the results, and would the results continue to hold if the assumptions are relaxed? In terms of the empirics, what are the implied assumptions underlying the causal argument? How valid are they? Etc.

**b. Term Paper (40%) –** This will take the form of a research proposal. There is no word-limit. It will be due close to the end of the semester (a specific date will be provided in class). It will address the following three points (at a minimum) -

**i.** What is the research question you are interested in addressing and why should one care about it? (**"Introduction"**)

**ii.** What are the closest papers that have considered similar issues? How will your question or approach differ from what they have done? (**"Literature Review"**)

**iii.** How do you intend to answer the question? (**"Analysis Overview"**).
Note that there is absolutely no restriction on how you intend to approach the problem (using a mathematical model (i.e. theory), empirical methods, or a combination of both). The analysis does not have to be complete. This means that you don't need to have results. However, the approach needs to be logically sound and achievable. If you want to do theory, then you need to provide the setup of the model, why do you think it would be the right model to use, how might one go ahead in terms of solving the model, and what would be the major roadblocks one might face? If you want to use empirical methods, then you need to mention what datasets and empirical strategies would be able to address the problem and make a convincing argument for the validity of the causal technique you wish to propose.

**c. Final Exam (40%) -** Comprehensive exam covering all topics taught in the course; to be held during the final exams week.

## 7. Attendance Policy:

Attendance will be marked using the Acadly app. There is no minimum attendance requirement; however, regular attendance is strongly encouraged, as active participation during class will help students better understand the material.

## 8. Recommended Books:

Detailed lecture notes will be provided. The course will cover seminal peer-reviewed journal articles in the field; a partial selection is listed below (***bold-italicized*** letters at the end of each article denote the relevant topic in the table above). The following textbooks, though not required, are very useful references. Dynamic programming methods to solve macro, and other commonly used techniques to solve micro and game-theoretic models in public economics are covered in MWG and LS. AS is a classic textbook, while G and RG provide less mathematically rigorous introductions to the topics covered in the course, particularly from an applied perspective. Students interested in reading further and doing research should look at AF, K and PT. SR is an excellent textbook that covers India's experience with fiscal federalism. Some of the material from the lectures will be directly based on chapters from these books (in particular LS, PT, SR, and K).

**a. Textbooks:**

1. Atkinson and Stiglitz (AS), *Lectures on Public Economics*, Princeton University Press
2. Auerbach and Feldstein (eds) (AF), *Handbook of Public Economics* Vols 1,2,3,4 & 5, Elsevier
3. Gruber (G), *Public Finance and Public Policy*, Macmillan UK
4. Kocherlakota (K), *New Dynamic Public Finance*, Princeton University Press
5. Ljungqvist and Sargent (LS), *Recursive Macroeconomic Theory*, The MIT Press
6. Mas-Colell, Whinston, and Green (MWG), *Microeconomic Theory*, Oxford University Press
7. Persson and Tabellini (PT), *Political Economics: Explaining Economic Policy*, The MIT Press
8. Rosen and Gayer (RG), *Public Finance*, McGraw-Hill
9. Singh and Rao (SR), *Political Economy of Federalism in India*, Oxford University Press (India)

**b. Journal articles:**

1. Arulampalam, W., Dasgupta, S., Dhillon, A., and B. Dutta (2009), "Electoral goals and center-state transfers: A theoretical model and empirical evidence from India", *Journal of Development Economics*, Vol. 88, No. 1, pp. 103-119 (***FF***)
2. Baron, D., and J. Ferejohn (1989), "Bargaining in legislatures", *American Political Science Review*, Vol. 83, No. 4, pp. 1181-1206 (***E&P***)

3. Barro, R. J. (1974), "Are government bonds net wealth?", *Journal of Political Economy*, Vol. 82, No. 6, pp. 1095-1117 (***DOT***)
4. Barro, R. J. (1979), "On the determination of the public debt", *Journal of Political Economy*, Vol. 87, No. 5, pp. 940-971 (***DOT***)
5. Battaglini, M., and S. Coate (2008), "A dynamic theory of public spending, taxation, and debt", *American Economic Review*, Vol. 98, No.1, pp. 201-236 (***DPP***)
6. Boadway, R., and M. Keen (1993), "Public goods, self-selection, and optimal income taxation", *International Economic Review*, Vol. 34, No. 3, pp. 463-478 (***PG***)
7. Chamley, C. (1986), "Optimal taxation of capital income in general equilibrium with infinite lives", *Econometrica*, Vol. 54, No. 3, pp. 607-622 (***DOT***)
8. Coate, S. (1995), "Altruism, the Samaritan's dilemma, and government transfer policy", *American Economic Review*, Vol. 85, No. 1, pp. 46-57 (***OTP***)
9. Das, P., and C. Ghate (2022), "Debt decomposition and the role of inflation: A security level analysis for India", *Economic Modelling*, Vol. 113 (***PDI***)
10. Diamond, P. A. (1975), "A many-person Ramsey tax rule", *Journal of Public Economics*, Vol. 4, No. 4, pp. 335-342 (***SOT***)
11. Diamond, P. A., and J. A. Mirrlees (1971), "Optimal taxation and public production I: Production efficiency", *American Economic Review*, Vol. 61, No. 1, pp. 8-27 (***SOT***)
12. Kanbur, R., and M. Keen (1993), "Jeux sans frontieres: Tax competition and tax coordination when countries differ in size", *American Economic Review*, Vol. 83, No. 4 pp. 877-892 (***FF***)
13. Kaplow, L. (2006), "Public goods and the distribution of income", *European Economic Review*, Vol. 50, No. 7, pp. 1627-1660 (***PG***)
14. Khemani, S. (2007), "Does delegation of fiscal policy to an independent agency make a difference? Evidence from intergovernmental transfers in India", *Journal of Development Economics*, Vol. 82, No. 2, pp. 464-484 (***FF***)
15. Lindbeck, A., and J. Weibull (1987), "Balanced-budget redistribution as the outcome of political competition", *Public Choice*, Vol. 52, No. 3, pp. 273-297 (***E&P***)
16. Lizzeri, A., and N. Persico (2001), "The provision of public goods under alternative electoral incentives", *American Economic Review*, Vol. 91, No. 1, pp. 225-239 (***E&P***)
17. Mirrlees, J. A. (1971), "An exploration in the theoryof optimum income taxation", *Review of Economic Studies*, Vol. 38, No. 2, pp. 175-208 (***SOT***)
18. Persson, T., an G. Tabellini (1999), "The size and scope of government: Comparative politics with rational politicians", *European Economic Review*, Vol. 43, pp. 699-735 (***E&P***)
19. Saez, E. (2002), "Optimal income transfer programs: Intensive versus extensive labor supply responses", *Quarterly Journal of Economics*, Vol. 117, No. 3, pp. 1039-1073 (***OTP***)
20. Samuelson, P. A. (1954), "The pure theory of public expenditure", *Review of Economics and Statistics*, Vol. 36, No. 4, pp. 387-389 (***PG***)
21. Sandmo, A. (1975), "Optimal taxation in the presence of externalities", *Swedish Journal of Economics*, Vol. 77, No. 1, pp. 86-98 (***E***)
22. Weitzman, M. L. (1974), "Prices versus quantities", *Review of Economic Studies*, Vol. 41, No. 4, pp. 477-491 (***E***)

**c. Other readings:**

1. Fiscal Responsibility and Budget Management Act (2003), *Ministry of Finance*, Government of India
2. Indian Public Finance Statistics (2019), *Dept. of Economic Affairs, Ministry of Finance*, Government of India
3. Medium Term Expenditure Statement (2023), *Ministry of Finance*, Government of India
4. State Finances: A Study of Budgets 2025-26 (2026), *Reserve Bank of India*
5. Union Budget (2025), *Ministry of Finance*, Government of India

## 9. Additional Information:

**a. Copyright:** The instructor of this course owns the copyright of all the course materials. The lecture materials are distributed only to the students attending the course (ECO731) at IIT Kanpur, and should not be distributed in print or through electronic media without the consent of the instructor. Students can make their own copies of the course material for their use

**b. Missing deadlines and late submission policy:** Students must provide a valid, documented excuse and notify the instructor in advance if they anticipate missing the specific assignment submission deadline. The validity of the excuse is entirely at the instructor's discretion. Each day beyond the deadline will incur a penalty of 5 points. For example, if the assignment, worth a total of 20 points, is due on September 7, and the student submits it on

September 9, their submission will be graded out of 10 points instead.

c. **Academic Misconduct:** This includes all forms of student academic misconduct wherever committed; illustrated by, but not limited to, cases of plagiarism and dishonest practices in connection with examinations. Any form of academic misconduct will lead to an automatic F in the course.

d. **Make-up exam policy:** A make-up exam option for the end-semester examination will be provided to students with valid grounds, in accordance with the Institute's rules.